



Supply Chain Logistics Intelligence

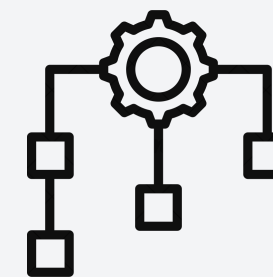
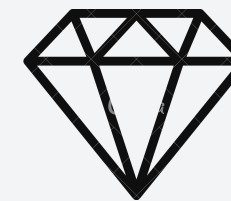
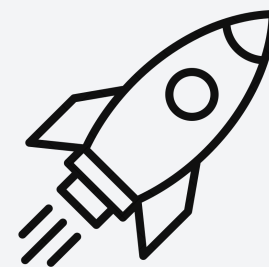
Don't stay raw, become the market law

Presented by: RAW2INSIGHT





Who We Are | RAW2INSIGHT



Our Identity

We are a data-driven accelerator dedicated to scaling startups and transforming them into industry leaders.

The Mission

We take scattered, raw logistics operations and engineer them into centralized, high-performance intelligence systems

The Value

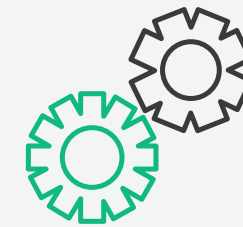
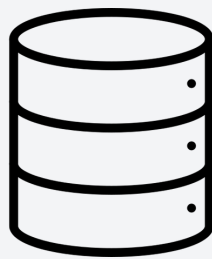
We don't just analyze numbers; we build the foundational data architecture that empowers startups to dominate their market.

Project Overview

This project transforms complex supply chain and logistics data into a clear, interactive intelligence dashboard. By analyzing key metrics like fuel costs, maintenance, and revenue, we empower decision-makers with actionable insights to optimize daily operations, cut costs, and maximize profitability.



Objectives



Centralize Data

Unify fragmented logistics data—including fuel, maintenance, and revenue—into a single reliable model.

Track Critical KPIs

Establish and monitor industry-standard metrics like RPM, CPM, and MPG to measure actual performance.

Optimize Costs

Identify spending patterns across different states and truck makes to reduce maintenance and fuel expenses.

Empower Decisions

Deliver a real-time, interactive dashboard that gives stakeholders complete visibility and strategic control.



Dataset Description



This dataset captures daily logistics operations, structured into three core areas:

- Financials: Freight revenue and customer billing.
- Operations: Routes, states, and delivery logs.
- Fleet Expenses: Fuel and maintenance costs by truck make.

This raw data is the foundation of our analytical model.

Data Preparation

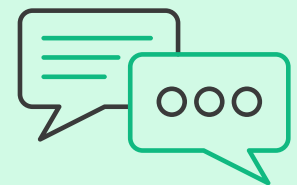
- Zero-Loss Cleaning: Handled nulls and fixed formats without deleting any historical records.
- Data Enrichment: Built custom DAX measures to track key metrics (RPM, CPM, MPG).
- Model Ready: Structured the raw logs perfectly for our relational Data Model.



Data Cleaning Preprocessing



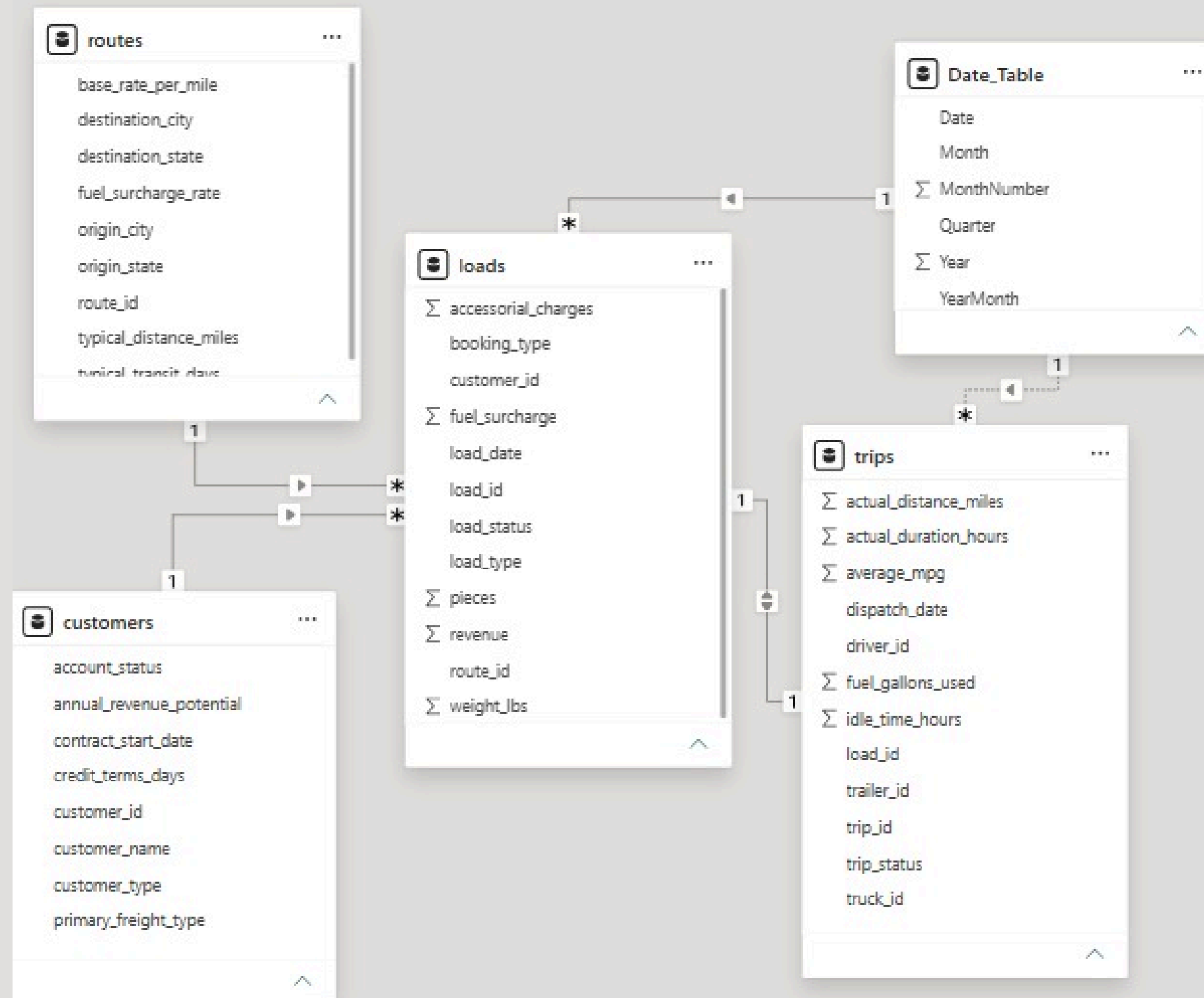
- Extracted and loaded 16 relational tables from the logistics database.
- Handled missing values and verified strict data types (e.g., Dates, Integers, Strings).
- Performed Outlier Detection using the IQR method.
- Calculated Data Skewness for numerical columns to ensure data quality.
- Standardized text formats and exported clean data for modeling.



Data Modeling in Power BI

- Imported clean datasets into Power BI.
- Designed a Star/Snowflake Schema.
- Created One-to-Many relationships between Dimension Tables (Customers, Drivers, Trucks) and Fact Tables (Loads, Delivery Events, Fuel Purchases).









Date_Table
Date
Month
Σ MonthNumber
Quarter
Σ Year
YearMonth

1



*

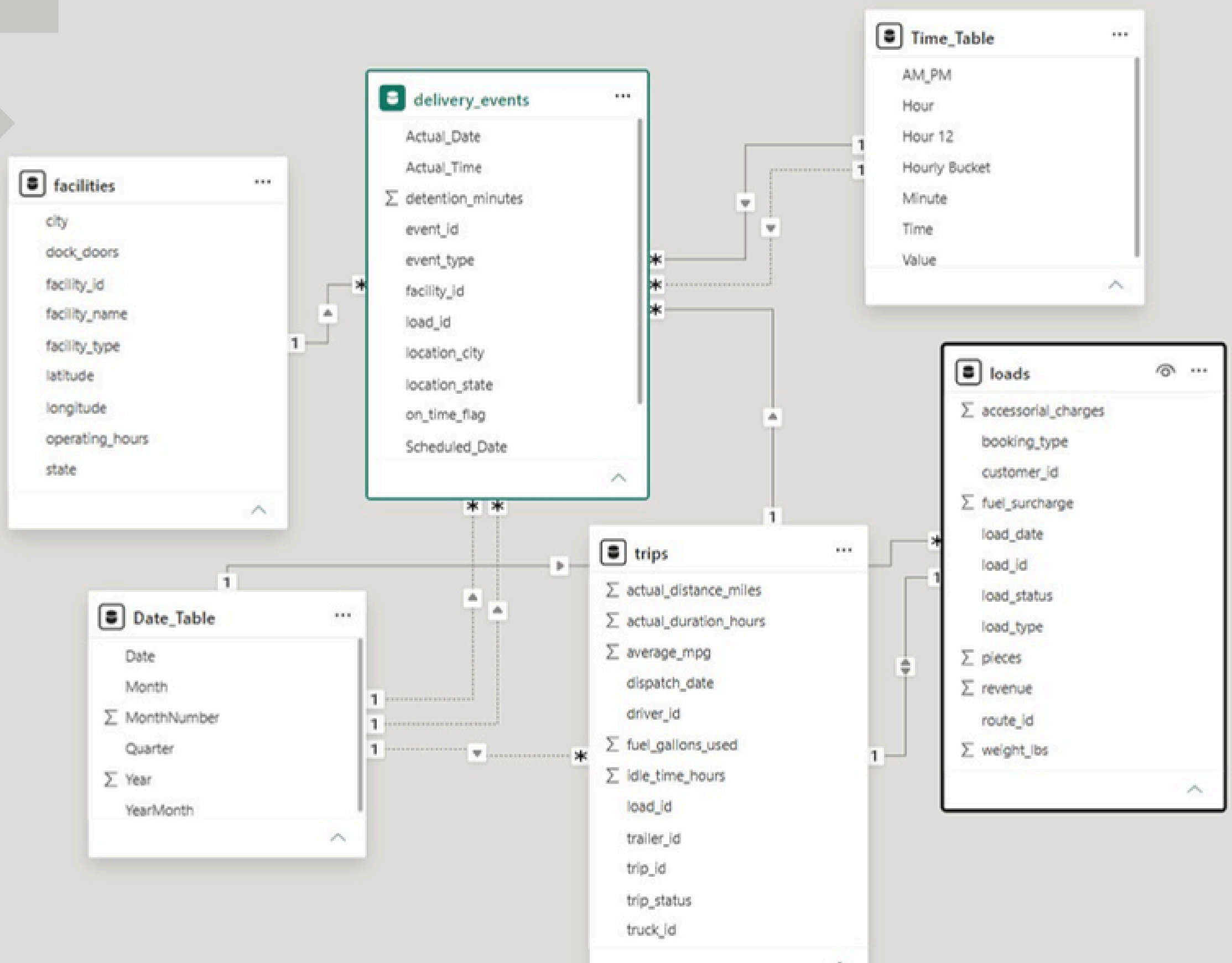
maintenance_records
Σ downtime_hours
facility_location
Σ labor_cost
Σ labor_hours
maintenance_date
maintenance_id
maintenance_type
Σ odometer_reading
Σ parts_cost
service_description
Σ total_cost
truck_id

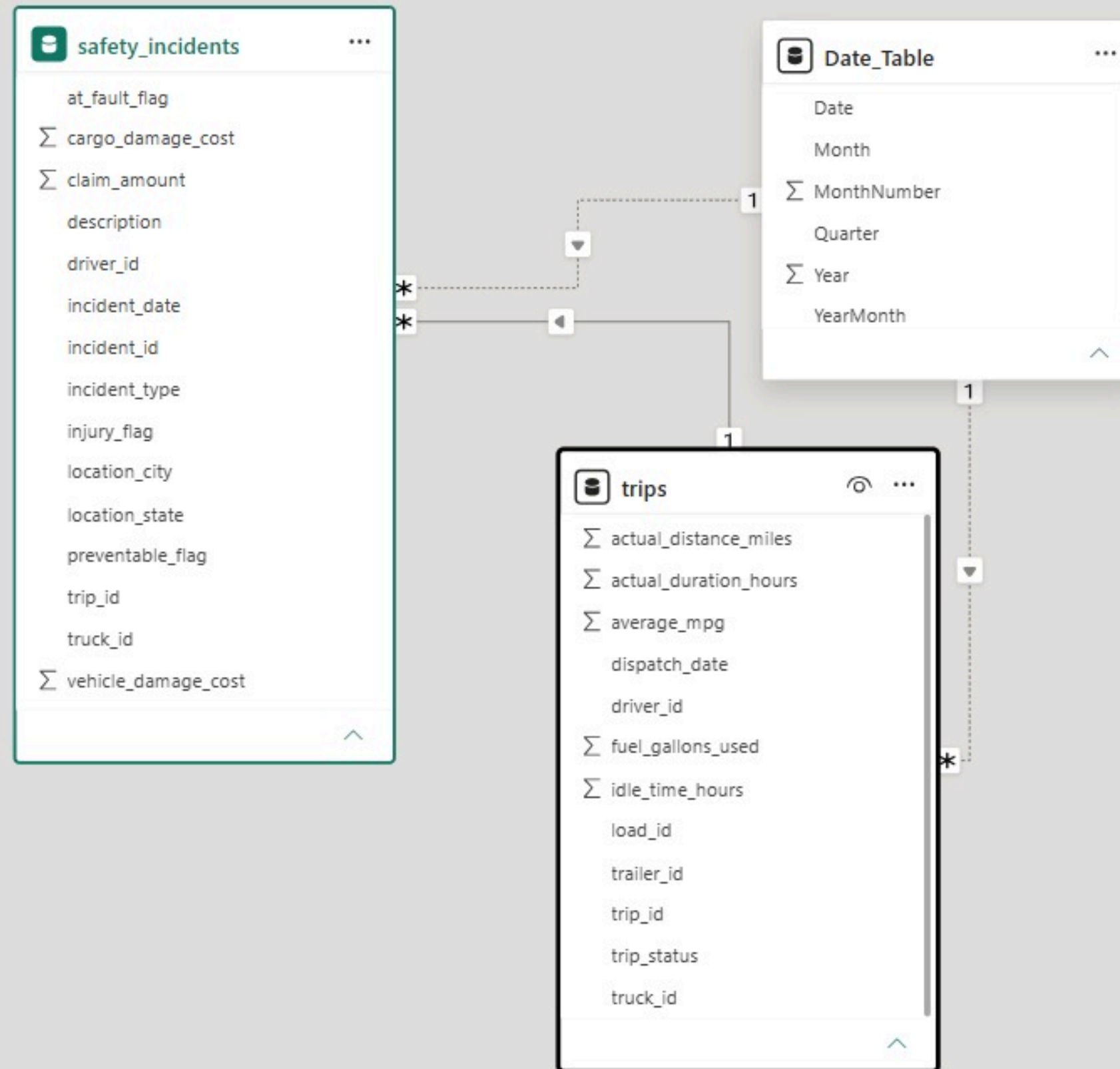
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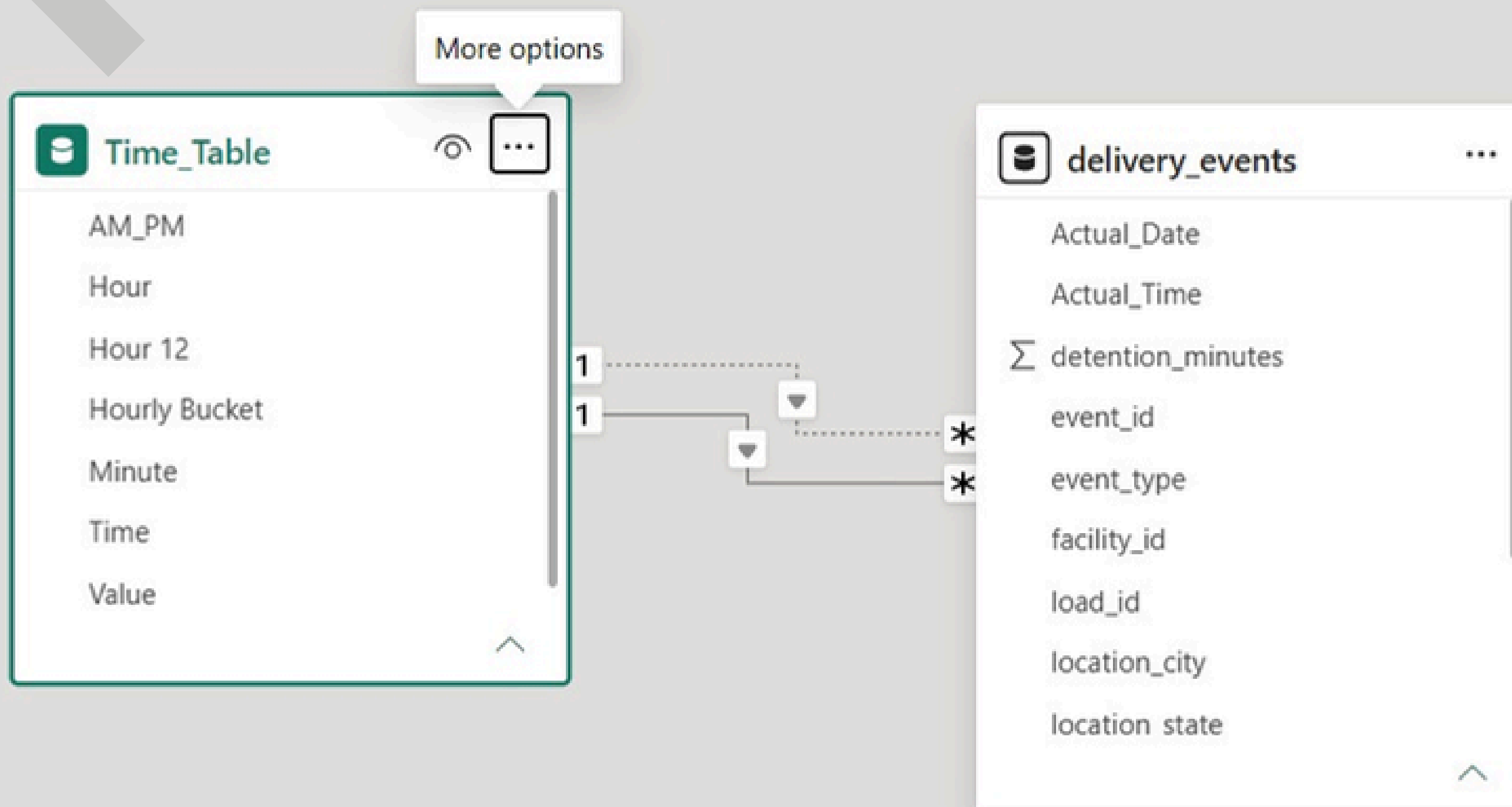
1

trucks
acquisition_date
Σ acquisition_mileage
fuel_type
home_terminal
make
Σ model_year
status
Σ tank_capacity_gallons
truck_id
Σ unit_number
vin









Operational Insights

This dashboard gives a full view of day-to-day fleet performance delivery reliability, idle time, load efficiency, and overall fleet utilization ,highlighting where operational gaps exist, where the fleet is performing well, and where seasonal patterns and regional bottlenecks are affecting results





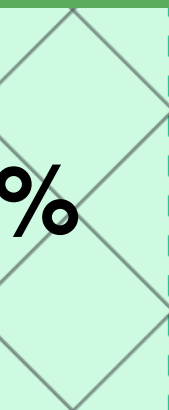
OTD rate: 44.62%

:Average Idle Hours
7.01

Average Load
Weight: 27.48K

:Miles per Tr
1.30M

relatively low, supposed to be in range of 90%-95%



:Average Idle Hours
7.01

Average Load
Weight: 27.48K

:Miles per Truck
1.30M

the current metric remains relatively high assuming the working hours are (12-10) hours a day



urs

Average Load
Weight: 27.48K

:Miles per Truck
1.30M

average, given the fact the general capacity is between 45k-48k about
58%-61%



ad
8K

:Miles per Truck
1.30M

An extremely high metric shows high productivity, although it might affect other factors like maintenance.

Count of load

Count of load_id by YearMonth



Count of Loads: Load volume relatively stayed stable 2023–2024, indicating consistent demand , also there’s was a noticeable increase during October and also an obvious setback during February. Recommendation: Maintain current operational capacity while focusing on improving efficiency and service quality rather than increasing volume . During peak-demand periods, the company should implement targeted marketing and customer engagement strategies to attract more customers and maximize revenue opportunities The October spike is most likely driven by the industry's peak shipping season (August–October), when retailers pre-stock inventory ahead of the holiday rush

Individual's idle time

- The average idle time was calculated utilizing aggregated data from all drivers. Overall, the current metric remains relatively high, indicating a critical need for operational review during the scheduling phase.



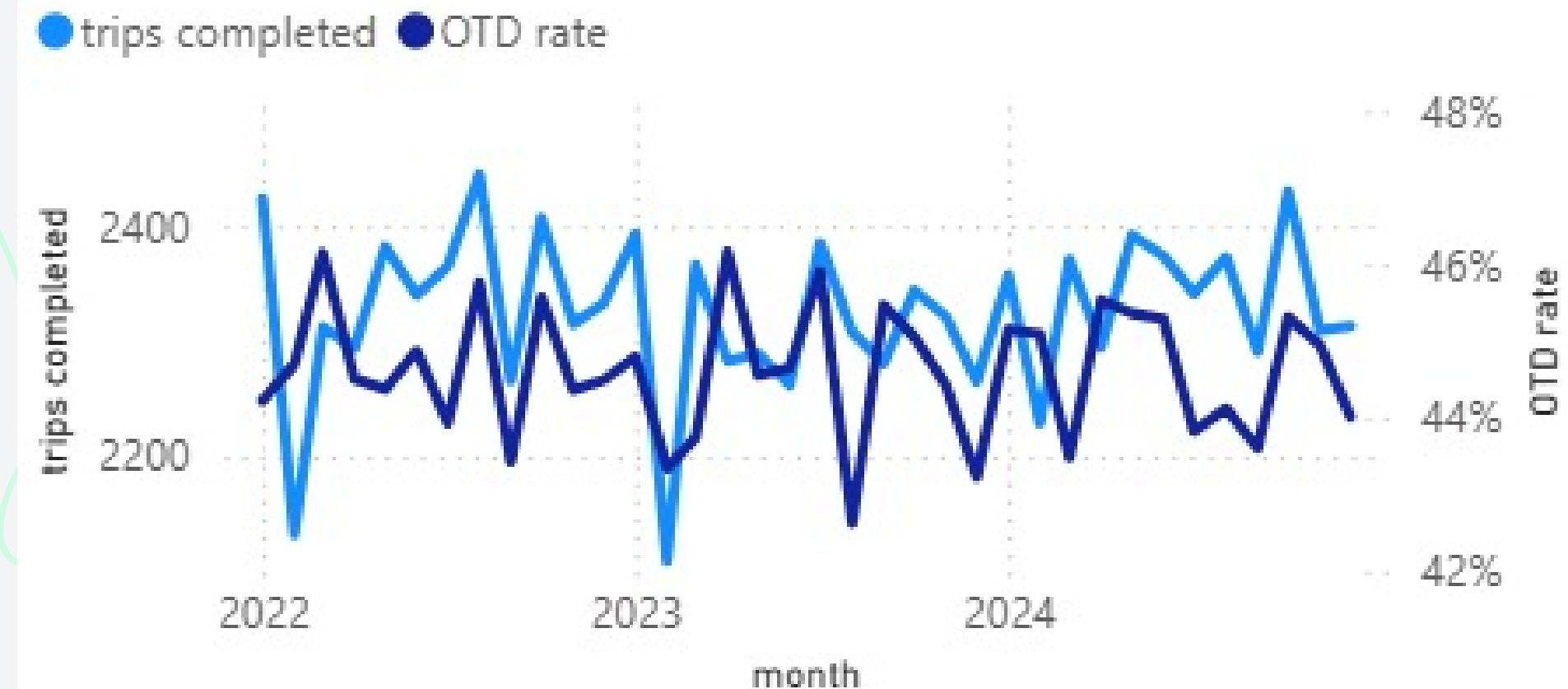
first_name	Average Idle Hours
David	7.07
William	7.06
Joseph	7.05
John	7.04
Patricia	7.04
Jennifer	7.03
Michael	7.03
Sarah	7.03
Robert	7.02
Total	7.01

OTD

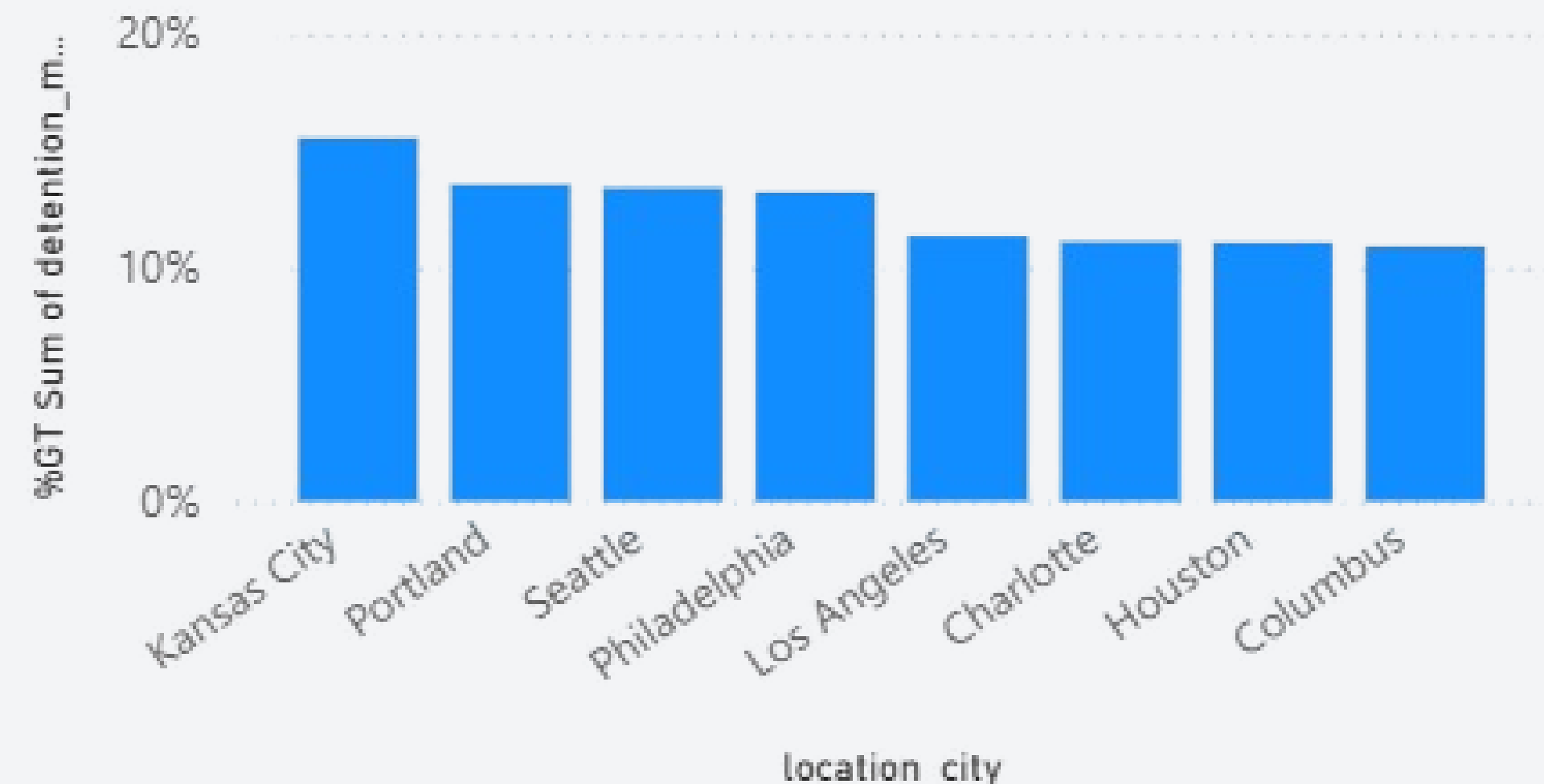
Analysis by TRIPS

- OTD remained relatively stable at 43–46% through 2024, which is well below the typical logistics industry benchmark of around 90–98% for on-time delivery. This indicates consistent underperformance in delivery reliability and suggests recurring operational constraints rather than isolated issues detention time by location, highlighting where delays are contributing to the low OTD rate.
- Recommendation:
- Focus on improving route planning, and try to manage detention time, and monitoring carrier performance. Prioritize to set incremental OTD improvement targets (e.g., 50% → 60% → 75%) while tracking performance monthly.

trips completed and OTD rate by month



%GT Sum of detention_minutes by location_city



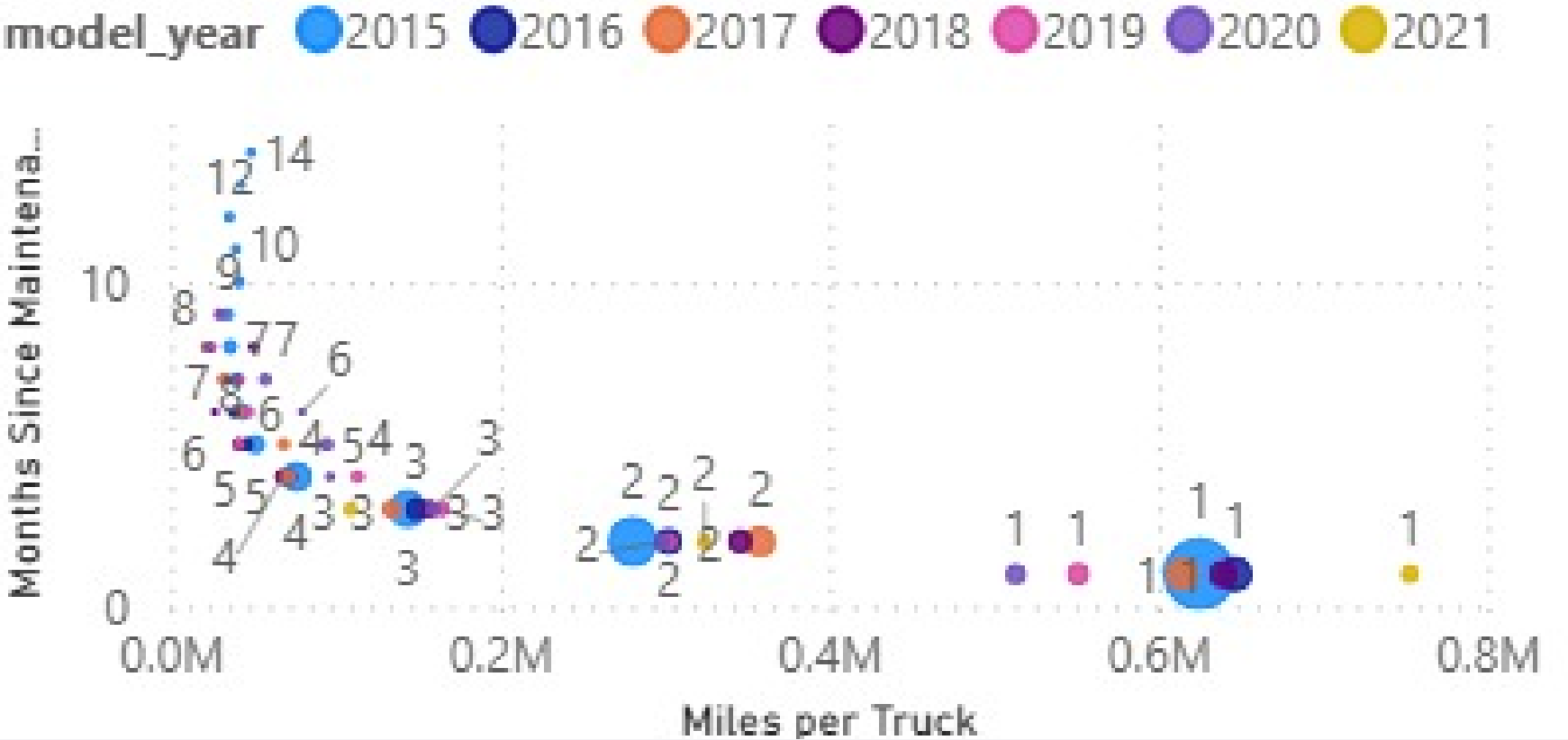
maintenance

Analysis by weight

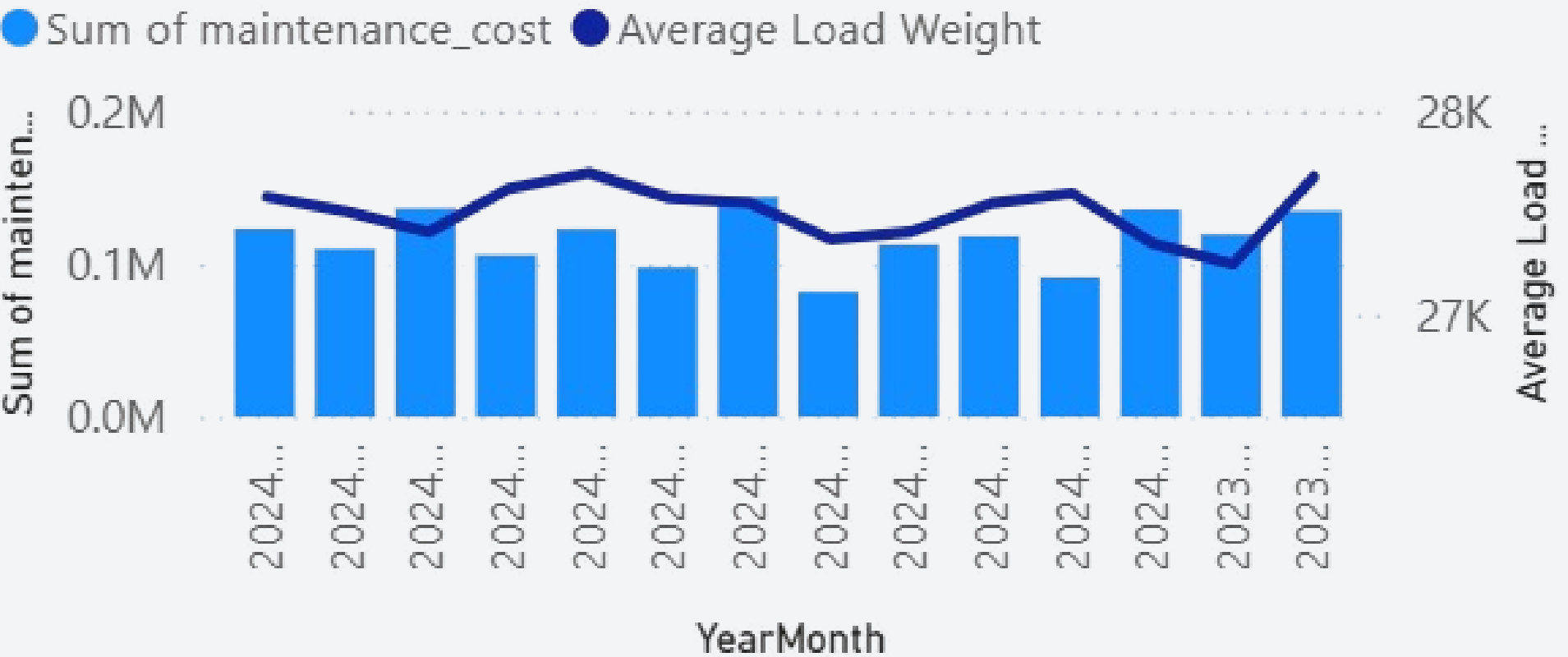
Monthly maintenance costs fluctuate unpredictably between \$0.05M and \$0.15M. While average load weight occasionally correlates with cost spikes, a distinct lag effect disrupts a perfect correlation. This lag represents a temporary "safety window" where newly repaired trucks briefly resist heavy payload stress before cumulative wear triggers the next breakdown. Crucially, this safety window is alarmingly short: 74% of maintenance expenses go to trucks re-entering the shop within 30 to 60 days. Notably, the aging 2013 models alone consume 33% of the fleet's total maintenance budget. This confirms that immediate failures stem from poor initial repair quality, heavily compounded by critical fleet depreciation rather than normal operational wear.

Recommendations: To eliminate rapid re-entry costs, implement strict workshop warranties and dynamically schedule preventive maintenance based on cumulative payload weight hauled. Additionally, collaborate with the Finance department to structure a phased installment plan for acquiring new trucks

Miles per Truck and %GT Sum of maintenance_cost by model_year and Months Since Maintenance



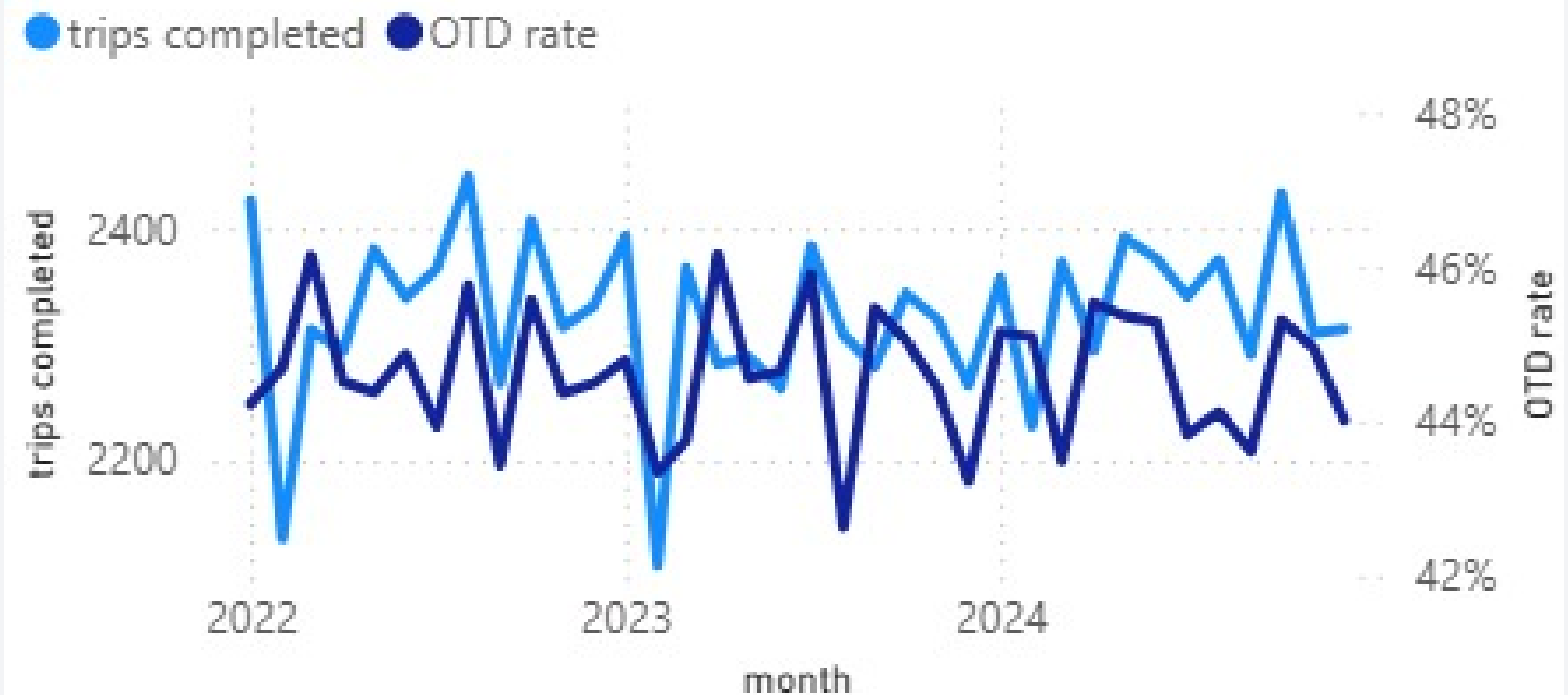
Sum of maintenance_cost and Average Load Weight by YearMonth



trips Analysis

- Monthly trips were generally stable with a peak at that frequently appear in
- October
- . This indicates consistent demand throughout the year, with a noticeable increase in activity during October throughout the year , however it's noticeable there's an obvious drop in February, in both cases this indict a seasonable change in demand in this period
- Recommendation: Since demand is generally stable with a peak in October and a decline in February, the company should prepare additional fleet and staffing resources for October while using the quieter February period for vehicle maintenance, staff training, and operational improvements

trips completed and OTD rate by month



Efficiency

Insights & Recommendations



Fleet Efficiency & Operational Cost Insights

**– Total Fuel Cost
\$95.59M**

**Total Maintenance
Cost – \$5.73M**

**CPM – Cost Per
Mile – \$0.83**

**Fuel Cost Per Mile
\$0.78 –**

The Primary Cost Driver

- Fuel expenditure reaches \$95.59M, heavily dominating our operational overhead.
- Represents the single largest financial burden and our primary target for immediate cost reduction.

Fleet Efficiency & Operational Cost Insights

**Total Maintenance
Cost – \$5.73M**

**CPM – Cost Per
Mile – \$0.83**

**Fuel Cost Per Mile
\$0.78 –**

**MPG – Miles Per
Gallon – 6.45**

Fuel vs. Maintenance Gap

- Total fleet maintenance stands at only \$5.73M compared to the \$95.59M fuel burden.
- Demonstrates that our main operational drain is fuel consumption, not mechanical repairs or asset upkeep.

Fleet Efficiency & Operational Cost Insights

**CPM – Cost Per
Mile – \$0.83**

**Fuel Cost Per Mile
\$0.78 –**

**MPG – Miles Per
Gallon – 6.45**

Total Operational Baseline

- Every mile driven by our fleet costs the company \$0.83 in combined operational expenses.
- This baseline CPM is the fundamental metric dictating our pricing strategy and profit margins per shipment.

Fleet Efficiency & Operational Cost Insights

– Fuel Cost Per Mile
\$0.78

MPG – Miles Per
Gallon – 6.45

The 94% Fuel Dominance

- Out of the \$0.83 total CPM, a staggering \$0.78 (approx. 94%) goes directly to fuel.
- Proves that our profitability per mile is almost entirely dictated by fuel consumption and market rates.

Fleet Efficiency & Operational Cost Insights

**MPG – Miles Per
Gallon – 6.45**

The Target for Optimization

- Our fleet currently operates at a baseline efficiency of 6.45 MPG.
- The Solution: Since fuel represents 94% of our CPM, even a fractional increase in MPG (via route optimization and idle reduction) will unlock millions in direct profit.

Geographic Cost Distribution

total_Cost by destination_state



- **High-Cost Outliers: Texas (TX) and Tennessee (TN)** show massive cost spikes, consuming the largest share of our operational budget.
- **Regional Concentration:** Expenses are heavily clustered across the Midwest and Southeast corridors, indicating regional routing bottlenecks.
- **Targeted Action:** Prioritize deploying fuel-efficient trucks (Kenworth) and AI dynamic routing specifically to these high-burn states.

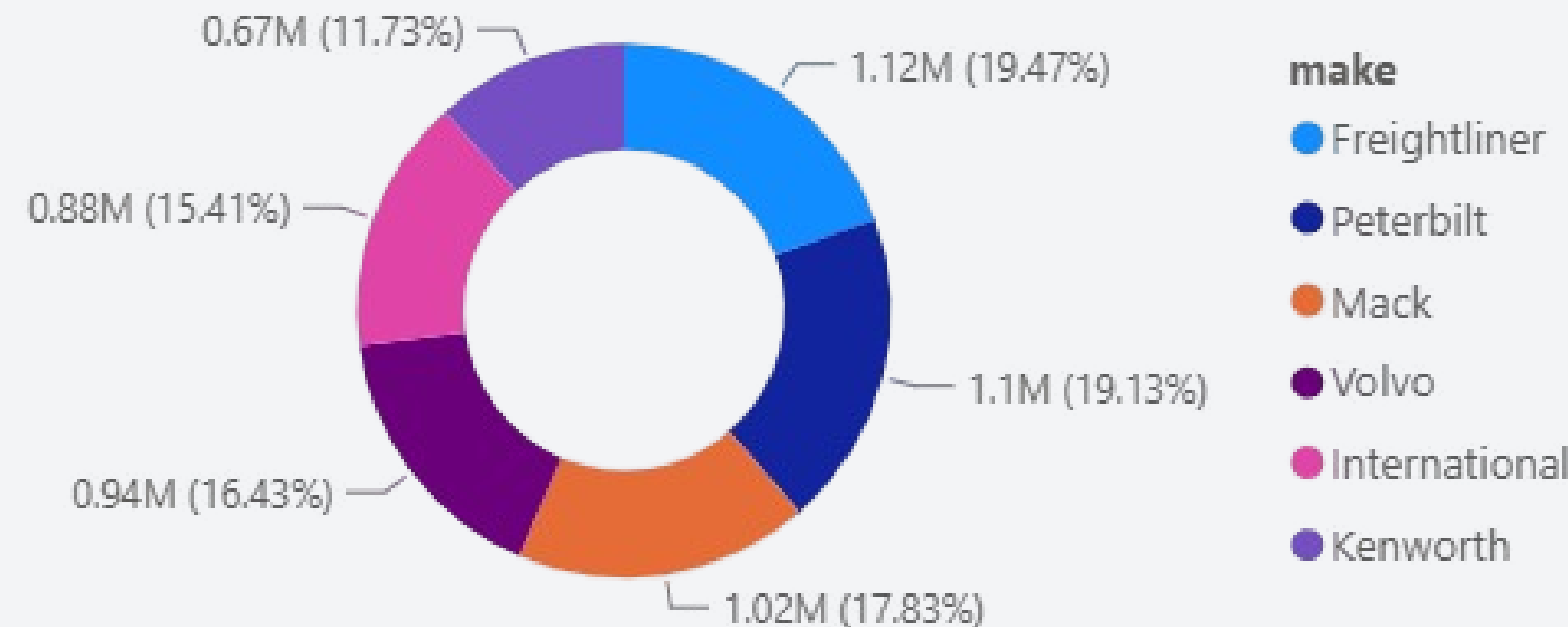
Fleet Maintenance

Analysis by Make

Maintenance Cost Breakdown

- Highest Maintenance Assets: Freightliner (\$1.12M | 19.47%) and Peterbilt (\$1.10M | 19.13%) represent our most expensive trucks to maintain, consuming nearly 38.6% of the total maintenance budget combined.
- The Cost-Effective Leader: Kenworth is by far the most economical asset, accounting for only 11.73% (\$0.67M) of total repairs—saving over \$450K compared to top-cost makes.
- Strategic Procurement: Future fleet replacement and purchasing strategies should shift away from Freightliner/Peterbilt and heavily prioritize Kenworth to minimize long-term overhead.

Total Maintenance Cost by make



Cost Anomaly

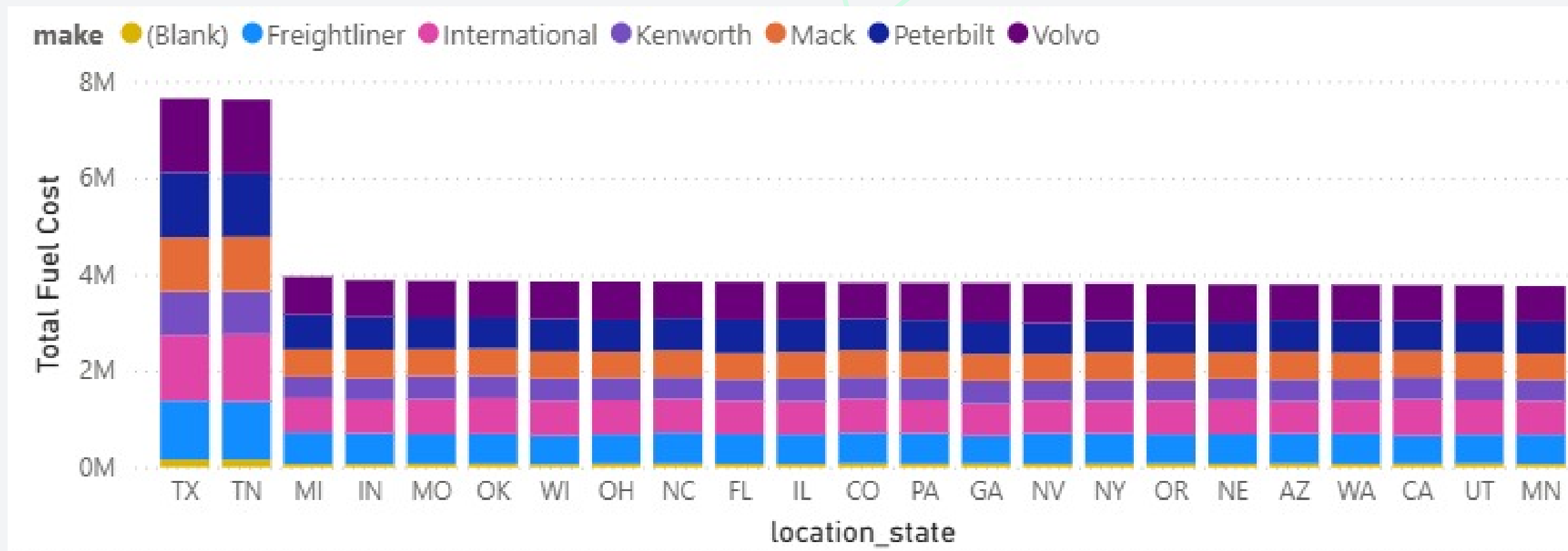
& Individual Asset Performance

Asset-Level Performance & Cost Spikes

- Age vs. Maintenance Proof: As shown in the table, older 2015 models (like TRK00083 and TRK00117) can operate at highly efficient maintenance costs (\$11.8K to \$25.5K).
- The Extreme Outliers: This proves that extreme spenders—like TRK00003 (\$90,161) and the \$70K+ anomaly group—are not failing just due to vehicle age, but due to severe mechanical defects or operational abuse.
- Targeted Audit: Instead of a blind fleet replacement, we must immediately isolate and decommission only these specific high-cost outliers to protect our \$5.73M repair budget.

truck_id	Total Maintenance Cost	model_year
TRK00083	11,896.28	2015
TRK00117	25,579.03	2015
TRK00007	25,868.97	2019
TRK00036	27,522.41	2020
TRK00047	28,243.39	2015
TRK00019	28,662.15	2015
TRK00076	30,133.21	2015
TRK00107	31,131.25	2017
TRK00101	32,866.18	2015
Total	5,730,573.28	

Geographic & Asset Fuel Breakdown



- TX & TN Spikes: Both states hit ~\$8M in fuel spend—double the ~\$4M average of all other states.
- Routing, Not Trucks: Truck brands are evenly split across all states, proving this cost spike is driven by regional operations/routing, not asset types.
- Action Plan: Prioritize AI dynamic routing and our most efficient trucks (Kenworth) in TX & TN to cut this \$16M+ drain.

Actionable Recommendations



- Targeted Route Optimization: Launch an immediate investigation and optimize routing specifically in Texas and Tennessee to reduce the severe fuel burn in these bottleneck states.
- Strategic Fleet Procurement: Shift future truck purchasing strategies towards Kenworth and away from Freightliner/Peterbilt to significantly reduce long-term maintenance overhead.
- MPG Improvement Initiatives: Since fuel dictates the CPM, implement driver efficiency training and reduce idle times to push the baseline 6.45 MPG higher, directly maximizing overall profit margins.

strategic dahsboard

**Insights &
Recommendations**



Profit per Load

2.33K

Gross Profit Margin

66.54%

Revenue per Mile

2.49

Total Revenue

298.62M

Revenue per Load

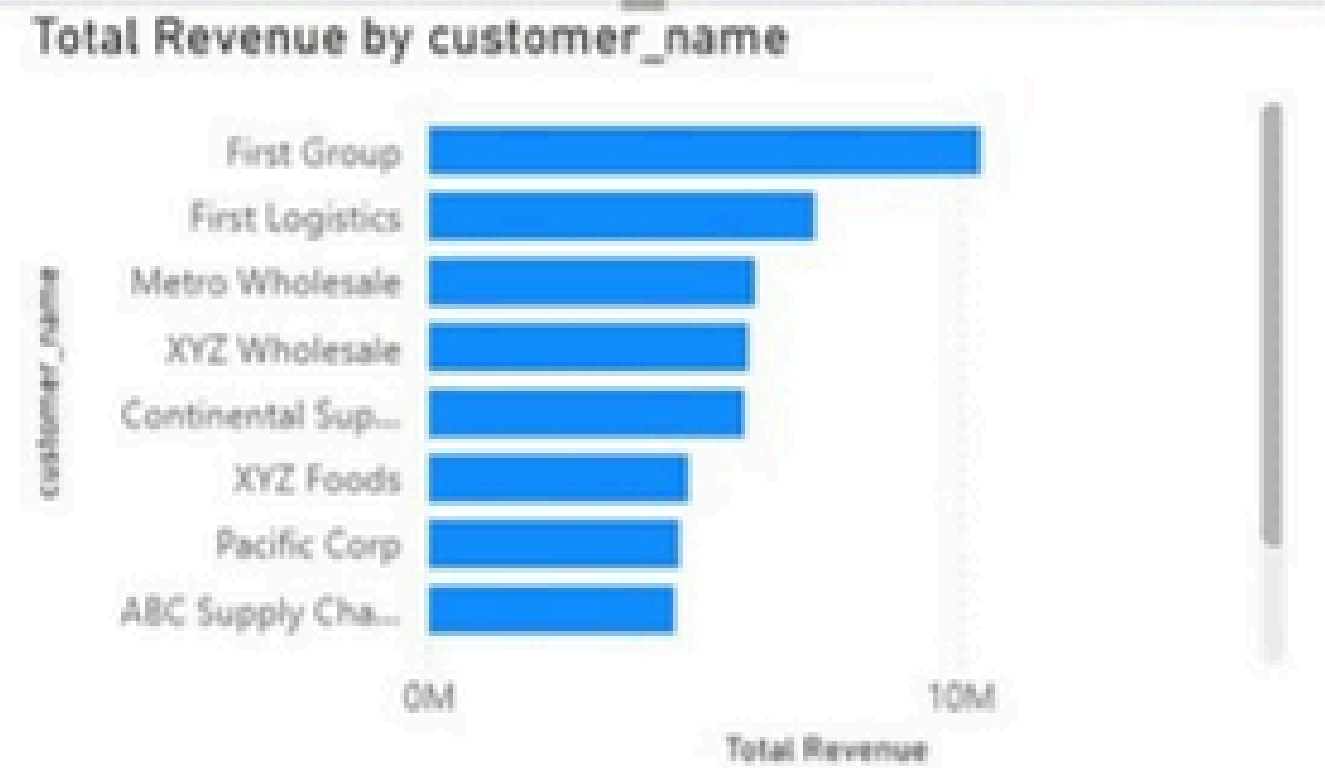
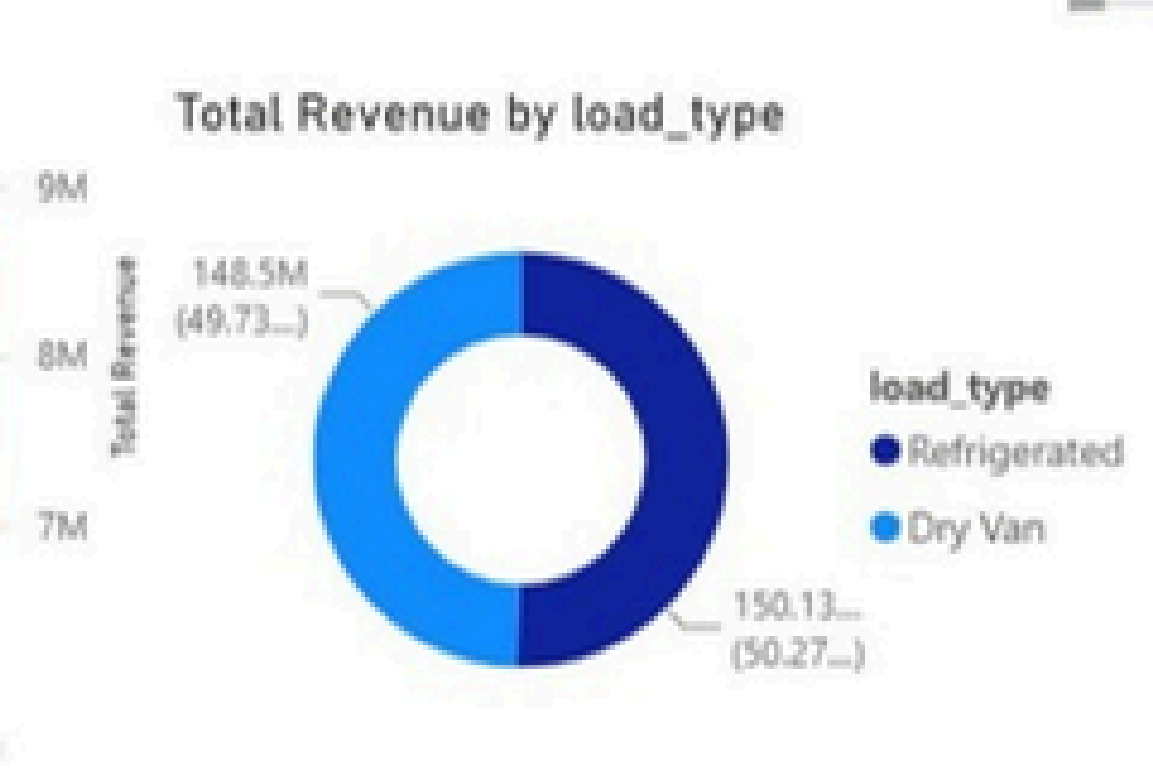
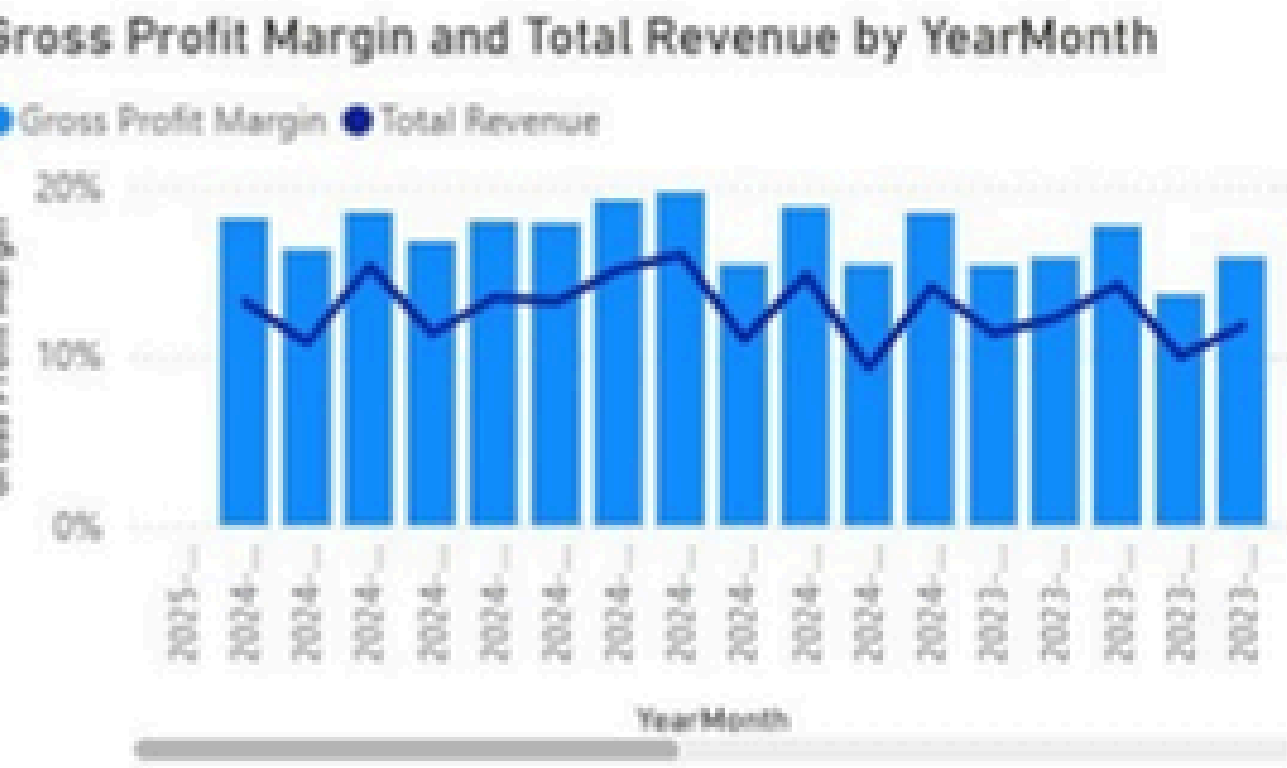
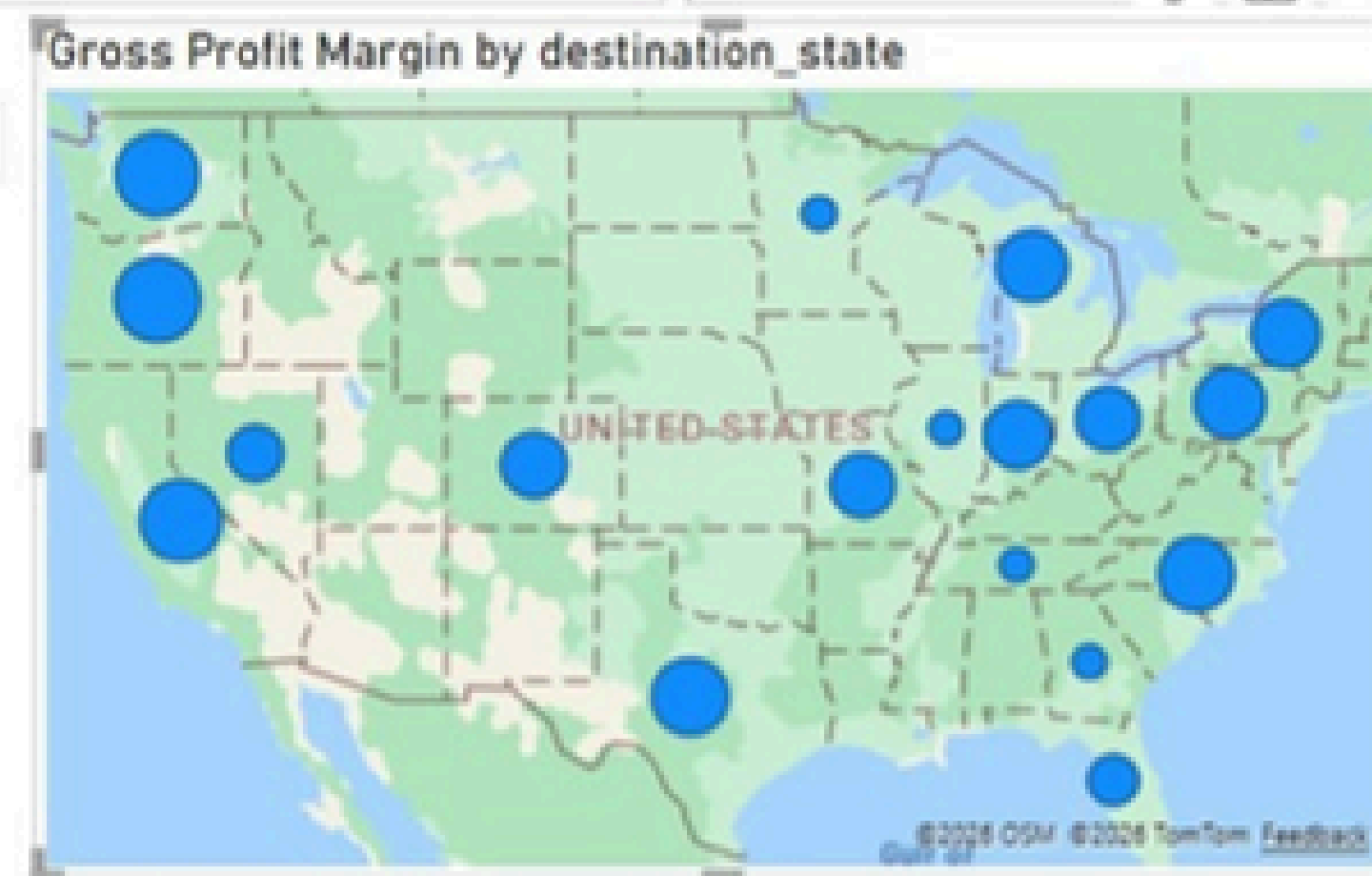
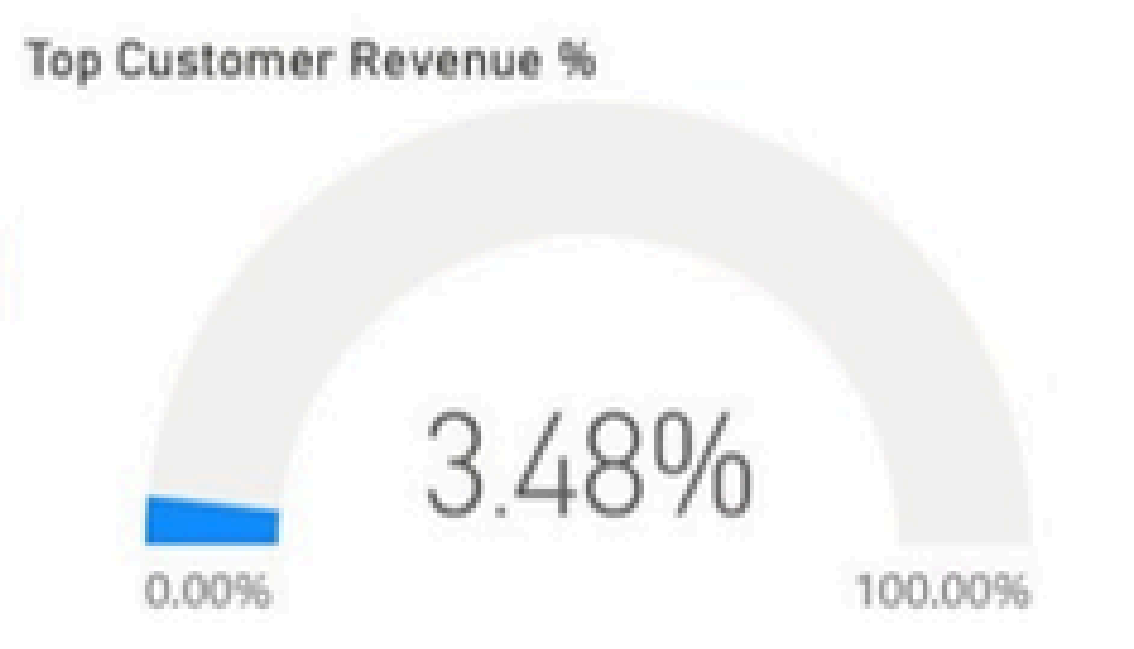
3.50K

Quarter

customer_type

All

All



Strategic Business Insights



- **Balanced Revenue Channels:** Dedicated fleets are the foundation, generating \$148.12M (49.6% of revenue). The rest is perfectly balanced between Spot Market (\$75.44M) and Contract accounts (\$75.06M).
- **Zero Concentration Risk:** The top customer accounts for only 3.48% of total revenue, ensuring financial safety and decentralized income.
- **Market Volatility Protection:** A perfect 50/50 hedge exists between Refrigerated Cargo (\$150.13M) and Dry Van Cargo (\$148.5M).

Actionable Recommendations



- Dedicated Fleet Optimization: Use uncommitted Spot capacity to fill empty backhaul legs on dedicated routes, pushing the RPM past \$2.49.
- Geographic Resource Allocation: Allocate additional equipment and trailer pools directly to the high-margin regional hubs identified on the map.
- Key Account Loyalty: Lock high-volume drivers (e.g., First Group, First Logistics) into long-term Dedicated agreements to secure predictable future revenue.

Data

Aggregation & Logic

- Data Integration: Consolidated trip, truck, and payload data to establish a centralized, unified database.
- Data Cleansing & Outlier Removal: Purged anomalous records to ensure data integrity (e.g., excluding trips with average speeds below 10 mph or exceeding 80 mph).
- Feature Engineering: Implemented driver-level monthly aggregation to evaluate performance trends over a sustained period, rather than assessing isolated trips.





Logistics Decision Intelligence

Professional AI-powered tools for supply chain operations.

Trip Segmentation (Clustering) Driver Efficiency Prediction

Trip Category Predictor

Distance (Miles)

500.00

- +

Weight (Lbs)

20000.00

- +

Duration (Hours)

10.00

- +

Idle Time (Hours)

2.00

- +

Predict Category

Unsupervised Learning: Trip Profiling

- Objective: Identify distinct trip patterns to optimize operational planning and resource allocation.
- Algorithm Applied: Utilized K-Means Clustering to segment trips into three primary profiles based on key metrics: distance, duration, payload weight, and idle time.
- Deployment Logic: The trained model is exported and integrated into the production pipeline. This enables automated, real-time classification of incoming trips, ensuring immediate and appropriate resource assignment.





Logistics Decision Intelligence

Professional AI-powered tools for supply chain operations.

Trip Segmentation (Clustering)

Driver Efficiency Prediction

Driver Efficiency Predictor

Predict estimated MPG based on input metrics.

Total Monthly Miles

5000

-

+

Average Load Weight

20000

-

+

Total Monthly Hours

150

-

+

Total Trips Count

20

-

+

Predict Efficiency

Supervised Learning: Fuel Estimation



- Objective: Accurately forecast the expected fuel consumption for each driver based on actual operational variables (e.g., distance, time, and volumetric weight).
- Algorithm Applied: Developed a robust Random Forest Regressor model, achieving an exceptional predictive accuracy of 98.9% (R-squared score).
- Performance Metrics: The Mean Absolute Error (MAE) indicates a highly reliable model, making it fully viable for deployment in a real-world production environment.



Production Logic & Efficiency Scoring

- Backend Architecture: The system ingests new operational data and processes it through the deployed model to instantly generate the "Expected Fuel Consumption."
- Efficiency Metric & Calculation: Driver performance is quantified using the actual MPG (Miles Per Gallon) metric. This is calculated strictly as: $\text{Actual Total Distance (Miles)} \div \text{Actual Fuel Consumed (Gallons)}$.
- Automated Evaluation: The system continuously benchmarks the actual consumption against the model's predictions. Drivers with significantly higher consumption than expected (low MPG) are automatically flagged for coaching and optimization, while highly efficient drivers are identified for recognition.



Future Strategic Roadmap



Future Strategic Roadmap



Phase 1: Near-Term Optimization (0–6 Months)

- Real-Time Dynamic Routing: Integrate live traffic and weather APIs to autonomously reroute trucks and eliminate regional bottlenecks in TX & TN.
- Expected Impact: Push baseline efficiency from 6.45 to 6.8+ MPG and reduce total idle time across the fleet.

Phase 2: Medium-Term Asset Control (6–18 Months)

- AI predictive Maintenance: Deploy machine learning sensors (IoT) on high-spend assets to forecast mechanical failures before breakdowns occur.
- Expected Impact: Stop emergency repair drains, directly lowering our \$5.73M maintenance overhead by eliminating extreme outliers (like TRK00003).

Phase 3: Long-Term Transition (18–36 Months)

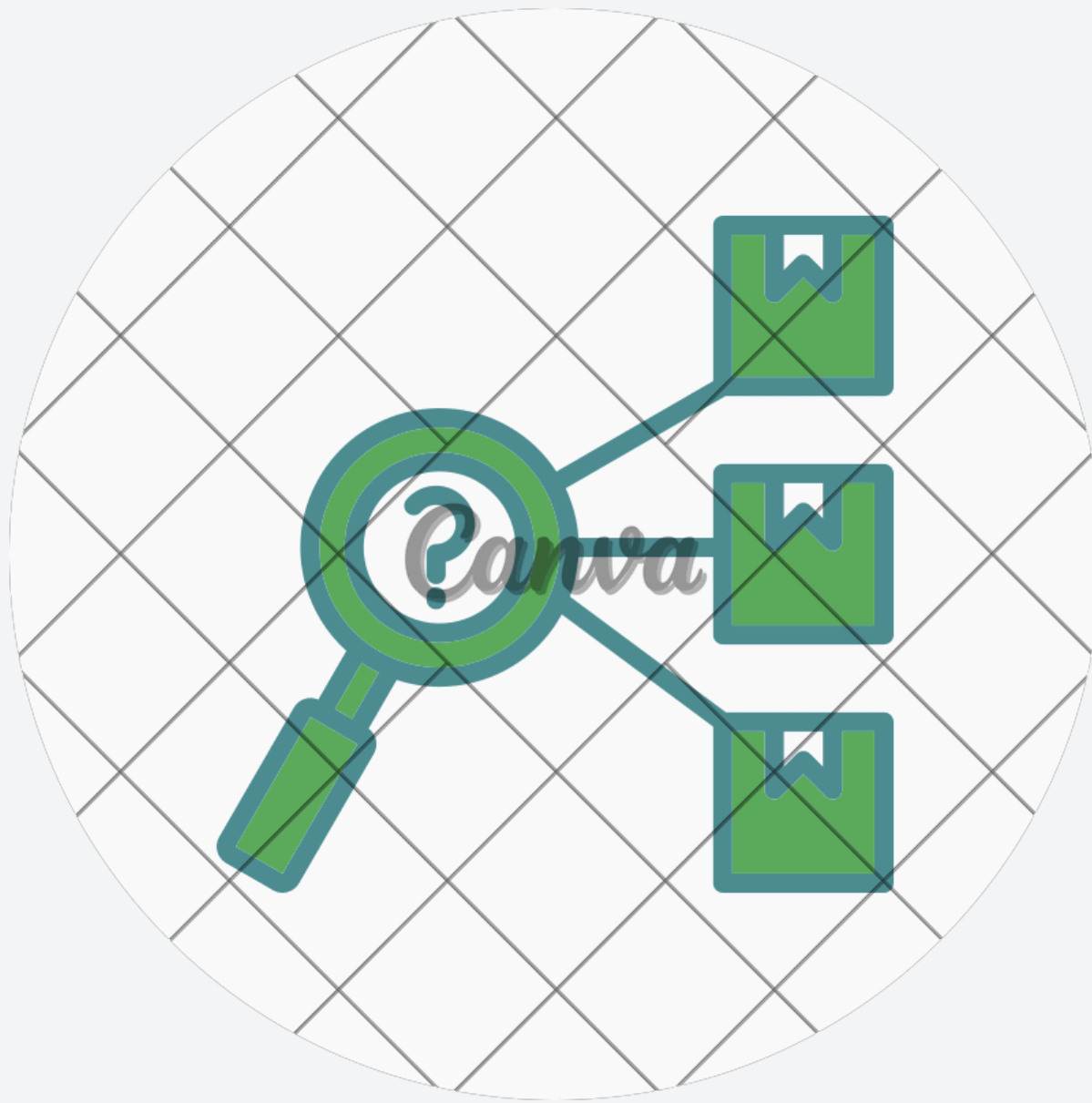
- Alternative Energy Fleet (EV / CNG): Pilot electric and compressed natural gas trucks specifically in high-burn regional corridors.
- Expected Impact: Aggressively shrink our massive \$95.59M fuel burden and decouple profit margins from volatile fossil fuel market prices.

Future Strategic Roadmap



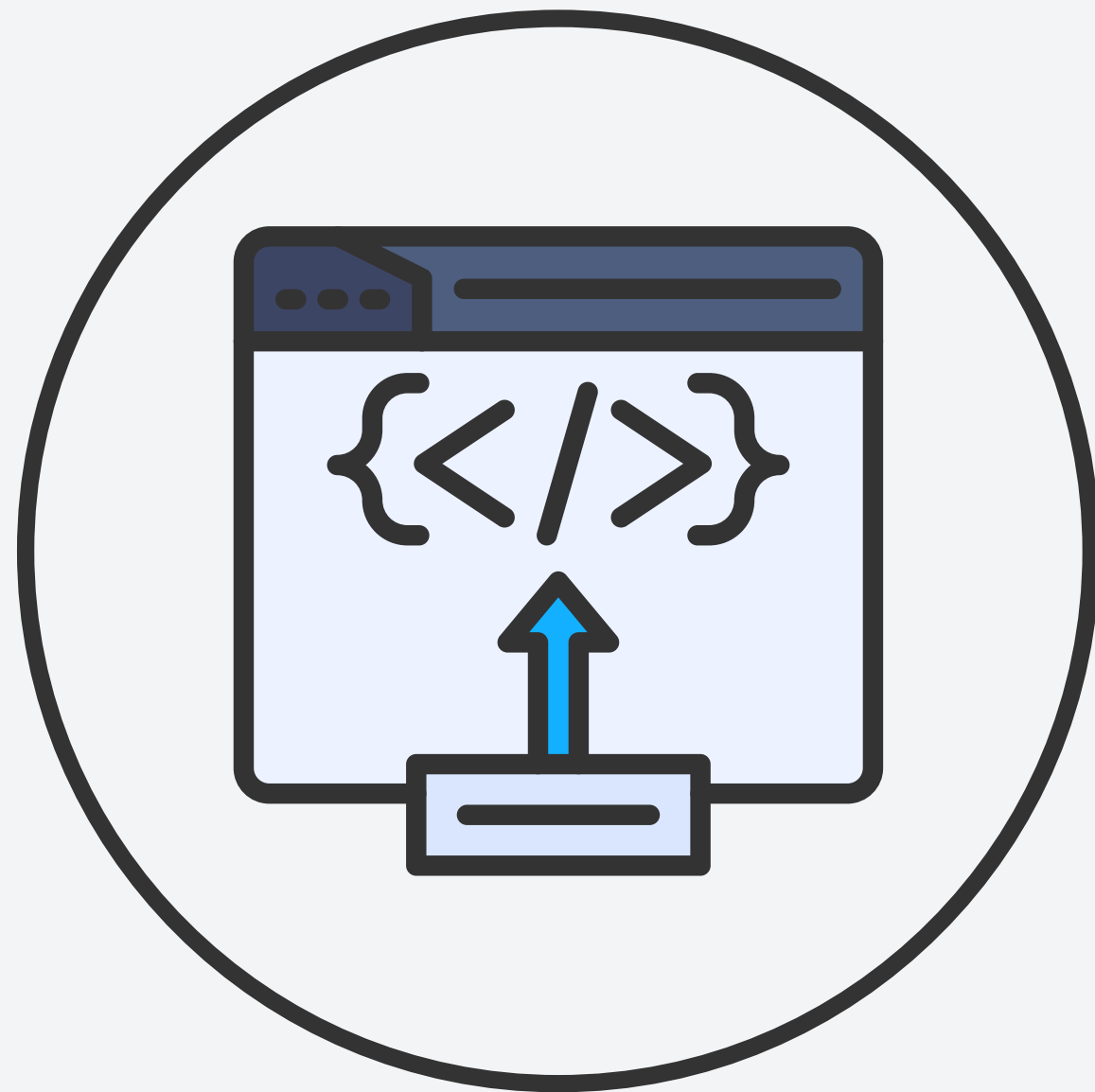
- Alternative Energy Transition (EV/CNG): Pilot electric or natural gas fleets in high-burn corridors (TX & TN) to aggressively shrink the \$95M fuel burden.
- AI-Driven Predictive Maintenance: Deploy machine learning to forecast breakdowns before they happen, eliminating costly downtime for high-maintenance assets.
- Real-Time Dynamic Routing: Integrate live traffic and weather APIs to autonomously reroute trucks, minimizing idle time and maximizing MPG.

References



- [1] U.S. Department of Energy (AFDC): Alternative Fuels Data Center - Heavy-Duty Vehicle Emissions and Cost Efficiency Metrics.
- McKinsey & Company (Supply Chain & Operations AI).
- Google Maps Platform - Routes & Traffic
- [5] Kenworth Truck Company: T680 & Fleet Efficiency Specifications (Fuel Economy and Lifecycle Maintenance Data).
- [4] American Transportation Research Institute (ATRI): An Analysis of the Operational Costs of Trucking (CPM & Maintenance Benchmarks).

deployment



Live End-to-End Solutions

- Interactive BI Dashboard (Web App): A fully automated, centralized tracking tool built to monitor fleet analytics, fuel consumption, and asset performance in real-time.
 - Live Link (Web App).
- Deep Learning AI Engine (Streamlit): An active predictive model deployed for category classification and automated decision-making across our operational pipelines.
 - Live Link: category-classifier123.streamlit.app



Thank You

Questions are welcome

